

LEFFIN

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New York City, NY

Engineer and creative technologist with 10+ years building software systems, automation infrastructure, and physical computing installations.

Background spans embedded hardware (Arduino, ESP32, Raspberry Pi, custom PCBs), Linux server administration, Python-heavy backend systems, parallel computing, and interactive art installations with computer vision.

Aerospace engineering foundation in optimization, physics, and computational methods.

PROFESSIONAL EXPERIENCE

Product Lead | Citibank

Dec 2024 – Present

AI-powered Internal Platform

- Architecting multi-agent AI system with intent resolution across UI state, data layers, and user context
- Designing system architecture and database schema for an agentic platform serving senior executives
- Defining orchestration patterns for real-time AI pipeline processing and content generation
- Collaborating directly with engineering team on implementation

Engineer & Data Scientist, Credit Risk Digital Tools | Citibank

Jul 2016 – Aug 2021

Automation systems, NLP tools, and large-scale data processing

- Built terminal-based automation tools using Python and Bash that reduced web accessibility audit time from 40 hours to under 5 minutes, processing all 72 WCAG guidelines using OCR and custom parsing
- Designed and deployed a parallel computing system in Python for automated letter review, parsing millions of documents daily across distributed processes
- Built an NLP-powered chatbot for regulatory policy traversal, using custom ML model to parse and query compliance documents via Excel front-end
- Developed AI system analyzing call center notes to identify customer complaints from unstructured text
- Built a decentralized ledger system using cryptographic hashing to manage strategy documentation
- Performed strategy code review for credit risk models using SAS for big data computation

Creative Technologist & New Media Artist

Aug 2021 – present

Interactive installations with physical computing, computer vision, and custom hardware

- Built real-time interactive installation using ML body segmentation and computer vision, optimizing complex rendering pipeline for responsive projection mapping (shown to 30,000+ at LUMA Festival)
- Designed and fabricated a 6x6x6 ft kinetic structure with custom PCB, integrating Arduino, sensors, and programmed interplay of light, movement, and sound (Grace Exhibition Space, NYC)
- Built AI-driven experience integrating four separate ML models (motion capture, voice synthesis, lip sync, dialogue generation) into a single real-time system with coherent state management
- Developed multi-state physical-digital installation using environmental sensors (mist, condensation) as triggers driving narrative state machine (selected for Currents New Media Festival 2026)
- Built 2D physics game with real-time generative AI object creation from natural language prompts, managing live physics simulation state

Graduate Research Assistant | Parsons School of Design

Oct 2021 – May 2023

- Multiple projects integrating AI/ML in creative coding using ml5.js, GPT-2 APIs, and custom ML pipelines
- Co-designed interactive quantum computing visualization in partnership with IBM using TouchDesigner

EDUCATION

Parsons School of Design | Masters in Design & Technology with Honors | 2021 - 2023

Relevant courses- Physical computing, machine learning, creative coding, interactive systems

IIT Bombay | Bachelors in Aerospace Engineering with Minor in Industrial Design | 2013 - 2017

Relevant courses- CUDA/parallel computing, optimization algorithms, numerical methods, computational fluid dynamics

SKILLS

Languages: Python, JavaScript/TypeScript, C# (Unity), SQL, Bash, HTML/CSS, Next.js/React

Systems & Infrastructure: Linux (Ubuntu Server, headless administration), Docker, self-hosted homelab, networking, SSH, systemd, Nginx

Embedded & Hardware: Arduino, ESP32, Raspberry Pi, custom PCB design and fabrication, sensors, actuators

Creative Technology: Computer vision (OpenCV, ML body segmentation), generative systems, Unity, p5.js

AI/ML & Data: NLP, ML model training, CUDA/GPU parallel computing, SAS (big data), data pipeline architecture

Math & Physics: Optimization algorithms, computational fluid dynamics, numerical methods, linear algebra

PERSONAL INFRASTRUCTURE

- Self-hosted Ubuntu Server homelab running multiple services including media servers, network management, and development environment
- Hand-built portfolio site (<https://leff.in>) in Next.js 16 with custom animation systems, deferred rendering, and complex state management

SELECT EXHIBITIONS

Currents New Media Festival	Santa Fe, NM	Jun 2026
NYCxDDesign “What Comes After No One”	New York, NY	May 2025
LUMA Festival	Binghamton, NY	Sep 2024
Parsons × LG AI Research “Art Design & AI”	New York, NY	Dec 2023
Pixelmouth “Follow the Machine”	New York, NY	Jun 2023
Parsons Thesis Show “Entropy”	New York, NY	May 2023
AMT Moving Image Festival	New York, NY	Apr 2023

MENTORSHIP & WORKSHOPS

Workshop, Physical Computing Workshop	The New School	2024
Mentor, Hardware Track, MIT Reality Hack	MIT Media Lab	2023
Mentor, Quantum Design Jam	IBM	2022